f1_simulator/data/alignment.py — Person 5 (Data alignment)

Takes one lap from telemetry.py (any source: OpenF1 or FastF1) and expresses it relative to the shared Track.

```python
from f1_simulator.core.track import Track
from f1_simulator.data.alignment import align_coordinates, convert_to_track_coo

track = Track()
lap_data = load_lap_csv("ver_monza_q.csv")          # Person 4's CSV: time,x,y,
aligned = align_coordinates(lap_data, track)         # x, y now in track metres
training_data = convert_to_track_coordinates(aligned, track)
print(alignment_report(training_data, track))
plot_alignment(aligned, track, "overlay.png")
```

Quick check (no repo needed)

```bash
pip install numpy scipy matplotlib pandas
python alignment.py      # self-test: prints PASS/FAIL, saves alignment_overlay
```

Inside the repo, put it at `f1_simulator/data/alignment.py` (add an empty `__init__.py` in `data/`); the self-test then also runs on Tony's `Track`.

Method

1. **Global fit (no initial guess):** resample the lap and the centreline by arc length; try every start offset, both directions and a mirrored frame; closed-form Procrustes (Umeyama) fit for each; keep the lowest residual.
2. **Refine:** ICP against the centreline using `track.project`, with robust (Huber) weights.
3. **Track coordinates:** sequential projection (falls back to a local search when the global projection jumps to another part of the circuit); heading from a smoothed trajectory; heading error wrapped to [-π, π).

Synthetic overlay

Synthetic test (first task) — `python -m pytest -q tests/test_alignment.py`

Known transform: scale 0.1 m/unit, rotation 0.7 rad, translation, random start point, noise. Recovered: scale 0.10002, rotation 0.701 rad; median s error 0.8 m; median lateral-offset error 0.7 m; 100% of samples within track widths; no backward steps. Mirrored frames are

detected when `allow_reflection=True` (off by default). Also passes on the team's `Track` (temporary ellipse).

Assumptions / limits

- The driver must travel in the direction of increasing track `s` (forward progress).
- Symmetric tracks such as the temporary ellipse match equally well rotated 180°, so `s` can be shifted by half a lap there; real circuits are not symmetric.
- Input must cover (nearly) a full lap — partial laps are not supported yet.
- The fit pulls the driver's line toward the centreline, so expect ~0.5–1 m bias in lateral offset. If Tony can give a known anchor (e.g., start/finish line position), we can remove most of it.
- Several laps from the same session share one frame: fit once, then pass `transform=` to `align_coordinates`.
- Only these Track members are used: `length, position, heading, left_width, right_width, project` (+ `boundaries` for the plot).